Velocity graphs and components

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Read a velocity graph and describe motion in two directions.

SUCCESS CRITERIA - I can...

1. Use signed velocity-time area to find displacement.
2. Represent horizontal and vertical components separately.
3. Distinguish graph height, slope and vector magnitude.

Core: teacher selects from 1-9. **Stretch:** 10 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use “component” and “axis” to explain your model or result. Sentence starters are optional.

component: the part of a vector along one chosen axis

axis: a reference line used to measure a direction

displacement: change from initial position to final position, with direction

01 Warm-up: bring back an earlier idea

A cart is at 4, 10, 16 m at 0, 1, 2 s.
Predict its position at 8 s.
State an assumption and a measurement that could challenge it.

Sentence starter: I predict … . My model supports this because … .

Marzano check - circle: ① ② ③

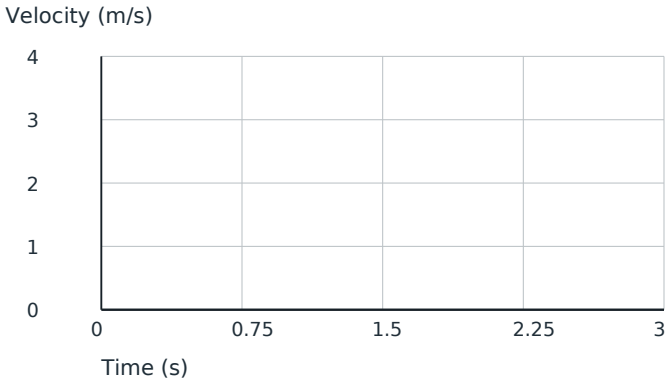
02 Worked example: explain the model

Reference: $\Delta x = v_x \Delta t$; $\Delta y = v_y \Delta t$.
For constant components. Subscripts x and y name the horizontal and vertical directions. Signed velocity-time area gives displacement.

Worked reasoning: A cart travels at 3 m/s east for 2 s. The rectangle under its velocity-time graph has area 3×2=6 m, its displacement.

Your turn: Draw the cart's constant velocity on the axes. Shade the area between the line and the time axis.
Explain why multiplying the two axis units gives meters.

Sentence starter: My claim is … . The diagram/data show …, so … .



Marzano check - circle: ① ② ③

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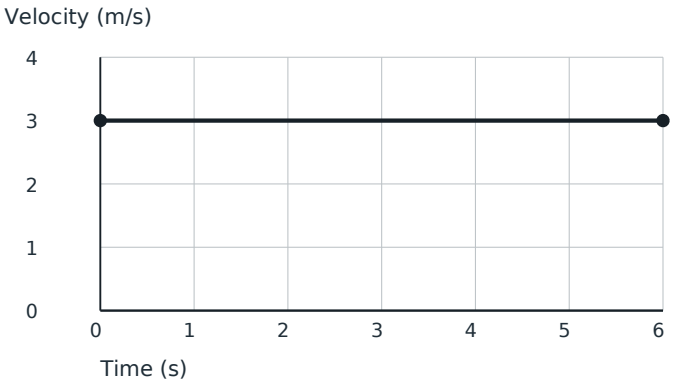
Continue down each column.

Sentence starters are optional. You may plan in any language.

03 Steady does not mean stopped

A remote-control car follows the velocity-time graph for 6 s. Is it stopped, moving steadily, or speeding up? Shade the area under the line and find its displacement.

Sentence starter: I found ... [unit]. This value means ... in this situation.

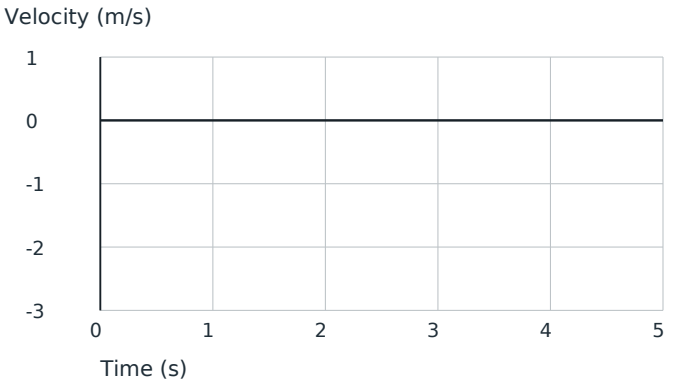


Marzano check - circle: 1 2 3

04 Move, then wait

A robot drives west at 2 m/s for 3 s, then waits for 2 s.
Draw its velocity-time graph, taking east as positive. How far west is it from the start?
Explain why waiting does not undo the trip.

Sentence starter: My claim is ... . The diagram/data show ..., so ... .

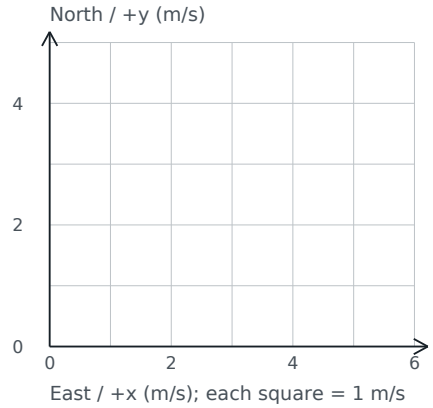


Marzano check - circle: 1 2 3

05 One diagonal motion

A drone moves 4 m east and 3 m north during each second of flight. These motions happen together.
Draw the velocity components on the grid and label each in m/s.

Sentence starter: The ... in my model represents ... . Its label shows ... .

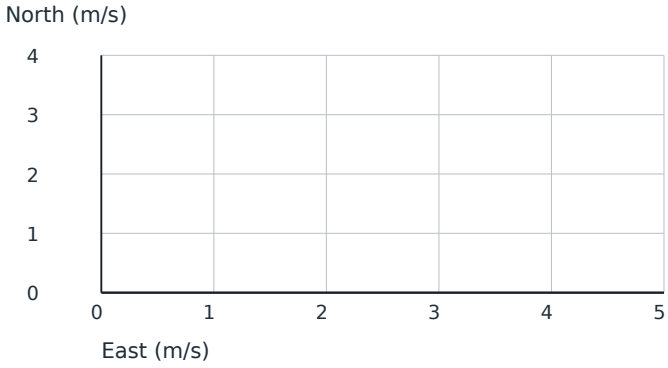


Marzano check - circle: 1 2 3

06 Make the components visible

For the drone in question 5, the diagonal velocity arrow has a supplied length of 5 m/s. A student adds 4 + 3 and reports 7 m/s.
Use the vector triangle to explain why that addition does not give the speed.

Sentence starter: My claim is ... . The diagram/data show ..., so ... .



Marzano check - circle: 1 2 3

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Continue down each column.

Sentence starters are optional. You may plan in any language.

07 Fix the speed claim

For a velocity with components 4 m/s east and 3 m/s north, the diagonal speed is given as 5 m/s. A student says it is 7 m/s.
Explain using a sketch.

Sentence starter: My claim is … . The diagram/data show …, so … .

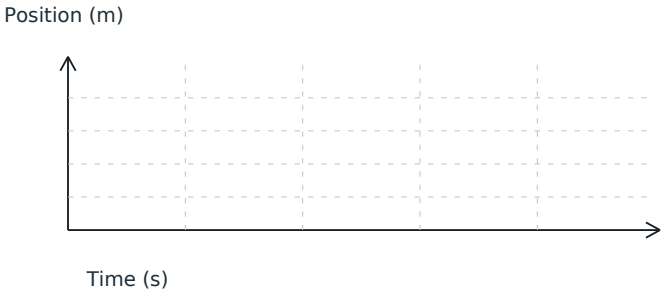
Marzano check - circle: 1 2 3

08 Review: connect an earlier idea

A cart goes from 15 m to 5 m in 3 s at steady velocity.
Find velocity and sketch its position-time graph.
Label the model with the given positions and any prediction.

Word help: velocity = rate of position change, with direction

Sentence starter: I predict … . My model supports this because … .



Marzano check - circle: 1 2 3

09 Exit check

A velocity graph stays at -2 m/s for 5 s.
Give displacement with direction. A different object has components 2 m/s east and 1 m/s south: sketch its motion direction.

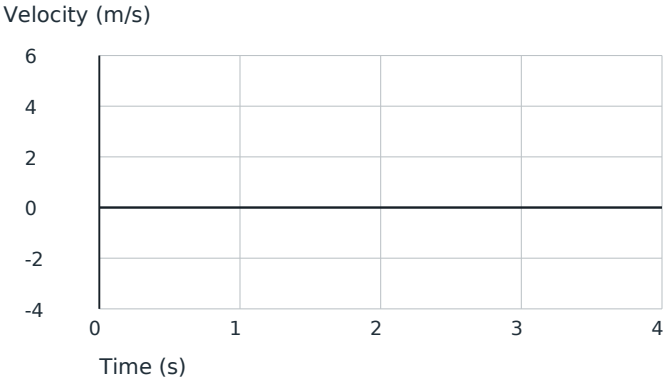
Sentence starter: Both … . They differ in … because … .

Marzano check - circle: 1 2 3

10 STRETCH: test the model

Draw a four-second velocity-time graph with nonzero motion but zero displacement.
Give its distance traveled.

Sentence starter: The … in my model represents … . Its label shows … .



Marzano check - circle: 1 2 3